
# Database Management System

# IT214

## Supply-Chain Management System *(SCMS)*

**Lab Group:** 5


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# Minimal FD Sets and Normalization Proofs (1NF → BCNF)

The following tables present the minimal (canonical) FD set for each relation. Trivial FDs are omitted. For surrogate-PK tables, the natural candidate key FD is always listed separately.

## SUPPLIER (Strong Entity)

Two candidate keys exist beyond the surrogate PK: supplier_code (business identifier), gstin (tax registration), and email — each unique by constraint. The generated column supplier_rating is determined by two non-key columns.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| supplier_id | → | {supplier_code, company_name, contact_person, email, phone, gstin, city, state, country, credit_limit, payment_terms_days, supplier_tier, on_time_delivery_rate, quality_rejection_rate, is_active, created_at, updated_at} | PK — single determinant for all non-key attributes |
| supplier_code | → | {supplier_id} (+ all other attributes via supplier_id) | Alternate CK — UNIQUE constraint |
| gstin | → | {supplier_id} (+ all other attributes via supplier_id) | Alternate CK — UNIQUE constraint |
| email | → | {supplier_id} (+ all other attributes via supplier_id) | Alternate CK — UNIQUE constraint |
| {on_time_delivery_rate, quality_rejection_rate} | → | supplier_rating | GENERATED ALWAYS AS STORED: ROUND((on_time × 0.6) + ((100 − rejection) × 0.4), 2). LHS is not a superkey — documented BCNF violation. |

Candidate Keys: {supplier_id}, {supplier_code}, {gstin}, {email}. Prime attributes: supplier_id, supplier_code, gstin, email.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values; no repeating groups; PK exists. | ✓**PASS** — All attribute domains contain scalar values. email and phone are real-world multivalued attributes but are simplified to single-valued columns (explicitly documented as a 1NF simplification in the schema). No arrays or nested structures exist. PK = supplier_id. 1NF holds. |
| 2NF | No partial dependency on any composite CK. | ✓**PASS** — Every CK in SUPPLIER is a single attribute. By definition, a single-attribute CK cannot have a proper subset, so no partial dependency is possible. 2NF is trivially satisfied. |

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 3NF | No non-prime attribute transitively depends on any CK. | **! Note** — All standard non-prime attributes depend directly on supplier_id with no intermediate non-prime attribute. However: {on_time_delivery_rate, quality_rejection_rate} → supplier_rating holds, where both source attributes are non-prime. Strictly, supplier_rating transitively depends on supplier_id via these two non-prime columns. The violation is mitigated in practice because supplier_rating is GENERATED ALWAYS AS STORED. The 3NF violation is thus a formal one with no practical consequence. Achieved 3NF practically (with documented caveat). |
| BCNF | Every non-trivial FD's LHS is a superkey. | × **FAIL** — Checking all non-trivial FDs: PK and alternates → {all} ✓(superkeys). Problematic FD: {on_time_delivery_rate, quality_rejection_rate} → supplier_rating, where LHS is NOT a superkey — strict BCNF violation. Retained here for KPI performance. Final verdict: 3NF achieved; BCNF violated (documented, justified). |

## PRODUCT_CATEGORY (Strong Entity — Recursive)

Two candidate keys: category_id (surrogate PK) and category_name (UNIQUE). The self-referential parent_category_id FK is an attribute, not a source of intra-table FDs beyond the PK.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| category_id | → | {category_name, parent_category_id, description} | PK |
| category_name | → | {category_id, parent_category_id, description} | Alternate CK — UNIQUE constraint |

Candidate Keys: {category_id}, {category_name}. Prime attributes: category_id, category_name.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values; no repeating groups; PK exists. | ✓**PASS** — All attributes are atomic scalars. Recursive hierarchy is modelled via a nullable self-referential FK. PK = category_id. 1NF holds. |
| 2NF | No partial dependency on any composite CK. | ✓**PASS** — Both CKs are single-attribute. 2NF trivially satisfied. |
| 3NF | No non-prime attribute transitively depends on any CK. | ✓**PASS** — Non-prime attributes: parent_category_id and description. parent_category_id is a self-referential FK. description is an independent attribute. No chain A → B → C exists within the table. 3NF holds. |
| BCNF | Every non-trivial FD's LHS is a superkey. | ✓**PASS** — Non-trivial FDs: category_id → {...} and category_name → {...} are both superkeys. BCNF holds. |

### PRODUCT (Strong Entity)

Two candidate keys: product_id (surrogate PK) and sku_code (UNIQUE). All product attributes describe a single product entity directly.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| product_id | → | {sku_code, product_name, category_id, unit_of_measure, is_perishable, shelf_life_days, standard_cost, reorder_point, reorder_quantity, weight_kg, is_active} | PK |
| sku_code | → | {product_id, product_name, category_id, unit_of_measure, is_perishable, shelf_life_days, standard_cost, reorder_point, reorder_quantity, weight_kg, is_active} | Alternate CK — UNIQUE constraint |

Candidate Keys: {product_id}, {sku_code}. Prime attributes: product_id, sku_code.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values; no repeating groups; PK exists. | ✓**PASS** — All attributes are scalar atomic types. No multivalued attributes or repeating groups. PK = product_id. 1NF holds. |
| 2NF | No partial dependency on any composite CK. | ✓**PASS** — Both CKs are single-attribute. 2NF trivially satisfied. |
| 3NF | No non-prime attribute transitively depends on any CK. | ✓**PASS** — is_perishable and shelf_life_days are conceptually related, but neither functionally determines the other. category_id is a FK reference. 3NF holds. |
| BCNF | Every non-trivial FD's LHS is a superkey. | ✓**PASS** — All determinants are superkeys. BCNF holds. |

### SUPPLIES (Associative Entity — M:N (Supplier × Product))

Surrogate PK supplier_product_id and natural candidate key {supplier_id, product_id, price_valid_from}. No partial dependency can exist because all non-key attributes describe the pricing contract for a specific supplier-product-period triple.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| supplier_product_id | → | {supplier_id, product_id, unit_price, currency, minimum_order_qty, lead_time_days, price_valid_from, price_valid_until, is_preferred_supplier} | Surrogate PK |
| {supplier_id, product_id, price_valid_from} | → | {supplier_product_id, unit_price, currency, minimum_order_qty, lead_time_days, price_valid_until, is_preferred_supplier} | Natural CK — UNIQUE constraint on triple |

Candidate Keys: {supplier_product_id}, {supplier_id, product_id, price_valid_from}. Prime attributes: supplier_product_id, supplier_id, product_id, price_valid_from.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values; no repeating groups; PK exists. | ✓**PASS** — All attributes are atomic scalars. PK = supplier_product_id. 1NF holds. |
| 2NF | No partial dependency on any composite CK. | ✓**PASS** — Checking the composite natural CK. All attributes (unit_price, currency, minimum_order_qty, etc.) depend on the full triple. No partial dependency. 2NF holds. |
| 3NF | No non-prime attribute transitively depends on any CK. | ✓**PASS** — None of the non-prime attributes determines another non-prime attribute. 3NF holds. |
| BCNF | Every non-trivial FD's LHS is a superkey. | ✓**PASS** — Non-trivial FDs are anchored on superkeys. BCNF holds. |

### WAREHOUSE (Strong Entity)

Two candidate keys: warehouse_id (surrogate PK) and warehouse_code (UNIQUE). All attributes describe warehouse properties directly.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| warehouse_id | → | {warehouse_code, warehouse_name, warehouse_type, city, state, pincode, total_capacity_sqft, temperature_controlled, is_active} | PK |
| warehouse_code | → | {warehouse_id, warehouse_name, warehouse_type, city, state, pincode, total_capacity_sqft, temperature_controlled, is_active} | Alternate CK — UNIQUE constraint |

Candidate Keys: {warehouse_id}, {warehouse_code}. Prime attributes: warehouse_id, warehouse_code.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values; no repeating groups; PK exists. | ✓**PASS** — All attributes are scalar atomic values. Physical address is decomposed. 1NF holds. |
| 2NF | No partial dependency on any composite CK. | ✓**PASS** — Both CKs are single-attribute. 2NF trivially satisfied. |
| 3NF | No non-prime attribute transitively depends on any CK. | ✓**PASS** — city → state could hold in real life, but not as an intra-relation FD enforcing consistency independent of warehouse_id. No other transitive chains. 3NF holds. |
| BCNF | Every non-trivial FD's LHS is a superkey. | ✓**PASS** — All determinants are superkeys. BCNF holds. |

### WAREHOUSE_ZONE (Weak Entity — owner: Warehouse)

Surrogate PK zone_id and composite natural CK {warehouse_id, zone_code}. zone_code alone is only a partial key (unique within a warehouse, not globally). All non-key attributes describe zone properties.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| zone_id | → | {warehouse_id, zone_code, zone_type, capacity_units, temperature_zone, is_active} | Surrogate PK |
| {warehouse_id, zone_code} | → | {zone_id, zone_type, capacity_units, temperature_zone, is_active} | Natural composite CK — UNIQUE(warehouse_id, zone_code) |

Candidate Keys: {zone_id}, {warehouse_id, zone_code}. Prime attributes: zone_id, warehouse_id, zone_code.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values; no repeating groups; PK exists. | ✓**PASS** — All attributes are scalar atomic types. 1NF holds. |
| 2NF | No partial dependency on any composite CK. | ✓**PASS** — zone_type does NOT depend on warehouse_id alone. All attributes require the full composite key. 2NF holds. |
| 3NF | No non-prime attribute transitively depends on any CK. | ✓**PASS** — temperature_zone and zone_type are correlated conceptually, but correlation is not functional dependency. 3NF holds. |
| BCNF | Every non-trivial FD's LHS is a superkey. | ✓**PASS** — All determinants are superkeys. BCNF holds. |

### BATCH (Strong Entity)

Two candidate keys: batch_id (surrogate PK) and batch_code (UNIQUE). All attributes describe a specific physical lot received from a supplier.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| batch_id | → | {batch_code, product_id, supplier_id, po_item_id, manufactured_date, expiry_date, received_date, initial_quantity, unit_of_measure, quality_status} | PK |
| batch_code | → | {batch_id, product_id, supplier_id, po_item_id, manufactured_date, expiry_date, received_date, initial_quantity, unit_of_measure, quality_status} | Alternate CK — UNIQUE constraint |

Candidate Keys: {batch_id}, {batch_code}. Prime attributes: batch_id, batch_code.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values; no repeating groups; PK exists. | ✓**PASS** — All attributes are scalar atomic values. 1NF holds. |
| 2NF | No partial dependency on any composite CK. | ✓**PASS** — Both CKs are single-attribute. 2NF trivially satisfied. |
| 3NF | No non-prime attribute transitively depends on any CK. | ✓**PASS** — supplier_id and product_id are direct properties of the batch. No non-prime attribute determines another. 3NF holds. |
| BCNF | Every non-trivial FD's LHS is a superkey. | ✓**PASS** — All determinants are superkeys. BCNF holds. |

### INVENTORY (Ternary Associative Entity — Product × Batch × Warehouse_Zone)

Surrogate PK inventory_id; natural CK {batch_id, zone_id}. Two additional FDs arise from design choices: batch_id → product_id (cross-table transitive redundancy) and the generated quantity_available.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| inventory_id | → | {product_id, batch_id, zone_id, quantity_on_hand, quantity_reserved, last_movement_at} | Surrogate PK |
| {batch_id, zone_id} | → | {inventory_id, product_id, quantity_on_hand, quantity_reserved, last_movement_at} | Natural CK — UNIQUE(batch_id, zone_id) |

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| batch_id | → | product_id | Cross-table transitive FD: Batch.product_id is already determined by batch_id in the BATCH table. Storing product_id here introduces redundancy — documented BCNF violation, retained for join-free query performance. |
| {quantity_on_hand, quantity_reserved} | → | quantity_available | GENERATED ALWAYS AS STORED: quantity_on_hand − quantity_reserved. LHS is not a superkey — additional normalization note. |

Candidate Keys: {inventory_id}, {batch_id, zone_id}. Prime attributes: inventory_id, batch_id, zone_id.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values; no repeating groups; PK exists. | ✓**PASS** — All attributes are scalar atomic types. 1NF holds. |
| 2NF | No partial dependency on any composite CK. | × **FAIL** — composite natural candidate key: {batch_id, zone_id}. The functional dependency batch_id → product_id exists. Since product_id (non-prime) depends on only a part of the candidate key (batch_id), this is a Partial Dependency. 2NF does not hold. |
| 3NF | No non-prime attribute transitively depends on any CK. | × **FAIL** — Transitive Dependency (violation): *inventory_id* → *batch_id* → *product_id*. {quantity_on_hand, quantity_reserved} → quantity_available is a 3NF concern but mitigated by GENERATED ALWAYS enforcement. 3NF fails. |
| BCNF | Every non-trivial FD's LHS is a superkey. | × **FAIL** — BCNF violation 1: batch_id → product_id. BCNF violation 2: {quantity_on_hand, quantity_reserved} → quantity_available. Final verdict: BCNF not achieved (both retained with justification). |

### INVENTORY_ALLOCATION (Associative Entity — SO_Item × Inventory)

Single surrogate PK allocation_id. No composite natural key enforced — each allocation event is a unique time-stamped record.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| allocation_id | → | {so_item_id, inventory_id, allocated_quantity, allocation_method, allocated_at, allocated_by} | PK — only determinant in this table |

Candidate Keys: {allocation_id}. Prime attributes: allocation_id.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values... | ✓**PASS** — 1NF holds. |
| 2NF | No partial dependency... | ✓**PASS** — Single-attribute PK. 2NF trivially satisfied. |
| 3NF | No transitive dependency... | ✓**PASS** — No non-prime determines another non-prime. 3NF holds. |
| BCNF | LHS is a superkey. | ✓**PASS** — BCNF holds. |

### PURCHASE_ORDER (Strong Entity)

Two candidate keys: po_id (surrogate PK) and po_number (UNIQUE business identifier). GRN fields are merged in from the eliminated GRN table; they are all fully determined by po_id (one GRN per PO, 1:1 relationship).

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| po_id | → | {po_number, supplier_id, warehouse_id, created_by, approved_by, po_status, order_date, expected_delivery_date, actual_delivery_date, received_by, grn_status, rejection_reason, payment_terms, payment_status, total_amount} | PK |
| po_number | → | {po_id} (+ all attributes via po_id) | Alternate CK — UNIQUE constraint |

Candidate Keys: {po_id}, {po_number}. Prime attributes: po_id, po_number.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values... | ✓**PASS** — 1NF holds. |
| 2NF | No partial dependency... | ✓**PASS** — Single-attribute CKs. 2NF holds. |
| 3NF | No transitive dependency... | ✓**PASS** — grn_status, received_by, etc., functionally depend on po_id, not each other. 3NF holds. |
| BCNF | LHS is a superkey. | ✓**PASS** — BCNF holds. |

### PO_ITEM (Weak Entity — owner: Purchase_Order)

Surrogate PK po_item_id; natural CK {po_id, product_id}. The stored line_total is computed from ordered_quantity × unit_price — introducing a functional dependency where the LHS is not a superkey.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| po_item_id | → | {po_id, product_id, supplier_product_id, ordered_quantity, received_quantity, unit_price, line_total, item_status} | Surrogate PK |
| {po_id, product_id} | → | {po_item_id, supplier_product_id, ordered_quantity, received_quantity, unit_price, line_total, item_status} | Natural CK — UNIQUE(po_id, product_id) |
| {ordered_quantity, unit_price} | → | line_total | GENERATED ALWAYS AS STORED: ordered_quantity × unit_price. This is an audit snapshot. LHS (ordered_quantity, unit_price) is not a superkey — BCNF violation, retained as price-at-order-time snapshot. |

Candidate Keys: {po_item_id}, {po_id, product_id}. Prime attributes: po_item_id, po_id, product_id.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values... | ✓**PASS** — 1NF holds. |
| 2NF | No partial dependency... | ✓**PASS** — No attribute depends on po_id or product_id alone. 2NF holds. |
| 3NF | No transitive dependency... | **! NOTE** — {ordered_quantity, unit_price} → line_total is a transitive dependency among non-prime attributes. Mitigated by GENERATED ALWAYS AS STORED. 3NF is technically (theoritically) violated but practically harmless. |
| BCNF | LHS is a superkey. | × **FAIL** — BCNF violation: {ordered_quantity, unit_price} → line_total. Retained as price-at-order-time snapshot. |

### CUSTOMER (Strong Entity)

Three candidate keys: customer_id (PK), customer_code (UNIQUE), and email (UNIQUE). outstanding_balance is NOT stored in the table — it is derived at query time via SUM over Sales_Order.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| customer_id | → | {customer_code, customer_type, company_name, contact_person, email, phone, billing_address, credit_limit, is_credit_hold, payment_terms_days, kyc_verified} | PK |

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| customer_code | → | {customer_id} (+ all attributes via customer_id) | Alternate CK — UNIQUE constraint |
| email | → | {customer_id} (+ all attributes via customer_id) | Alternate CK — UNIQUE constraint |

Candidate Keys: {customer_id}, {customer_code}, {email}. Prime attributes: customer_id, customer_code, email.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values... | ✓**PASS** — outstanding_balance is NOT stored. 1NF holds. |
| 2NF | No partial dependency... | ✓**PASS** — Single-attribute CKs. 2NF holds. |
| 3NF | No transitive dependency... | ✓**PASS** — credit_limit and is_credit_hold are correlated but not FD. 3NF holds. |
| BCNF | LHS is a superkey. | ✓**PASS** — BCNF holds. |

### SALES_ORDER (Strong Entity)

Two candidate keys: so_id (surrogate PK) and so_number (UNIQUE). Destination address is stored as decomposed flat fields.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| so_id | → | {so_number, customer_id, warehouse_id, created_by, so_status, order_date, requested_delivery_date, priority_level, dest_street, dest_city, dest_state, total_amount, payment_status} | PK |
| so_number | → | {so_id} (+ all attributes via so_id) | Alternate CK — UNIQUE constraint |

Candidate Keys: {so_id}, {so_number}. Prime attributes: so_id, so_number.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values... | ✓**PASS** — Delivery address is decomposed. 1NF holds. |
| 2NF | No partial dependency... | ✓**PASS** — Single-attribute CKs. 2NF holds. |
| 3NF | No transitive dependency... | ✓**PASS** — No non-prime → non-prime chain. 3NF holds. |
| BCNF | LHS is a superkey. | ✓**PASS** — BCNF holds. |

### SO_ITEM (Weak Entity — owner: Sales_Order)

Surrogate PK so_item_id; natural CK {so_id, product_id}. Parallel structure to PO_ITEM: line_total is stored as an audit snapshot introducing the same BCNF consideration.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| so_item_id | → | {so_id, product_id, ordered_quantity, allocated_quantity, picked_quantity, shipped_quantity, unit_price, line_total, item_status} | Surrogate PK |
| {so_id, product_id} | → | {so_item_id, ordered_quantity, allocated_quantity, picked_quantity, shipped_quantity, unit_price, line_total, item_status} | Natural CK — UNIQUE(so_id, product_id) |
| {ordered_quantity, unit_price} | → | line_total | GENERATED ALWAYS AS STORED: ordered_quantity × unit_price. Audit snapshot — same analysis as PO_ITEM. BCNF violation retained with justification. |

Candidate Keys: {so_item_id}, {so_id, product_id}. Prime attributes: so_item_id, so_id, product_id.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values... | ✓**PASS** — 1NF holds. |
| 2NF | No partial dependency... | ✓**PASS** — 2NF holds. |
| 3NF | No transitive dependency... | × **NOTE** — Same analysis as PO_ITEM: generated line_total. 3NF technically violated. |
| BCNF | LHS is a superkey. | × **FAIL** — BCNF violation retained with justification. |

### SHIPMENT (Strong Entity)

Two candidate keys: shipment_id (surrogate PK) and shipment_number (UNIQUE).

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| shipment_id | → | {shipment_number, origin_warehouse_id, vehicle_registration, carrier_name, driver_id, shipment_status, dispatch_at, estimated_arrival, actual_arrival, total_weight_kg, total_volume_cubic_m, freight_cost} | PK |
| shipment_number | → | {shipment_id} (+ all attributes via shipment_id) | Alternate CK — UNIQUE constraint |

Candidate Keys: {shipment_id}, {shipment_number}. Prime attributes: shipment_id, shipment_number.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values... | ✓**PASS** — Everything is decomposed. 1NF holds. |
| 2NF | No partial dependency... | ✓**PASS** — Single-attribute CKs. 2NF holds. |
| 3NF | No transitive dependency... | ✓**PASS** — No non-prime → non-prime chain. 3NF holds. |
| BCNF | LHS is a superkey. | ✓**PASS** — BCNF holds. |

### DRIVER (Strong Entity)

One candidate key: driver_id (PK).

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| driver_name | → | {driver_phone, driver_name} | PK |

Candidate Keys: {driver_id}. Prime attributes: driver_id.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values... | ✓**PASS** — Everything is decomposed. 1NF holds. |
| 2NF | No partial dependency... | ✓**PASS** — Single-attribute CKs. 2NF holds. |
| 3NF | No transitive dependency... | ✓**PASS** — No non-prime → non-prime chain. 3NF holds. |
| BCNF | LHS is a superkey. | ✓**PASS** — BCNF holds. |

### SHIPMENT_ITEM (Weak Entity — owner: Shipment)

Surrogate PK shipment_item_id; natural CK {shipment_id, so_item_id}. Minimal table with only three payload columns.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| shipment_item_id | → | {shipment_id, so_item_id, quantity_shipped} | Surrogate PK |
| {shipment_id, so_item_id} | → | {shipment_item_id, quantity_shipped} | Natural CK — UNIQUE(shipment_id, so_item_id) |

Candidate Keys: {shipment_item_id}, {shipment_id, so_item_id}. Prime attributes: shipment_item_id, shipment_id, so_item_id.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values... | ✓**PASS** — 1NF holds. |
| 2NF | No partial dependency... | ✓**PASS** — 2NF holds. |
| 3NF | No transitive dependency... | ✓**PASS** — Single non-prime attribute. 3NF trivially holds. |
| BCNF | LHS is a superkey. | ✓**PASS** — BCNF holds. |

### RETURN_REQUEST (Strong Entity)

Surrogate PK return_id. Contains a deliberate denormalization: customer_id is stored directly even though it is transitively determined via return_id → so_id → customer_id. This is a genuine 3NF (and BCNF) violation.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| return_id | → | {so_id, customer_id, approved_by, return_reason, return_status, requested_at} | PK |
| so_id | → | customer_id | Cross-table transitive FD: so_id → customer_id holds because Sales_Order.customer_id is determined by so_id. Storing customer_id here creates a transitive dependency within RETURN_REQUEST — 3NF violation, retained as deliberate denormalization for direct customer-level query access. |

Candidate Keys: {return_id}. Prime attributes: return_id.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values... | ✓**PASS** — 1NF holds. |
| 2NF | No partial dependency... | ✓**PASS** — Single-attribute PK. 2NF holds. |
| 3NF | No transitive dependency... | × **FAIL** — so_id → customer_id holds. return_id → so_id → customer_id. 3NF violated, retained as deliberate denormalization. |
| BCNF | LHS is a superkey. | × **FAIL** — BCNF is not achieved. |

### RETURN_ITEM (Weak Entity — owner: Return_Request)

Surrogate PK return_item_id; natural CK {return_id, so_item_id}. restocked_to_inventory_id is a nullable FK — it does not introduce intra-table FDs beyond the PK.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| return_item_id | → | {return_id, so_item_id, returned_quantity, condition_on_arrival, restocked_to_inventory_id, disposal_method} | Surrogate PK |
| {return_id, so_item_id} | → | {return_item_id, returned_quantity, condition_on_arrival, restocked_to_inventory_id, disposal_method} | Natural CK — UNIQUE(return_id, so_item_id) |

Candidate Keys: {return_item_id}, {return_id, so_item_id}. Prime attributes: return_item_id, return_id, so_item_id.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values... | ✓**PASS** — 1NF holds. |
| 2NF | No partial dependency... | ✓**PASS** — 2NF holds. |
| 3NF | No transitive dependency... | ✓**PASS** — 3NF holds. |
| BCNF | LHS is a superkey. | ✓**PASS** — BCNF holds. |

### STAFF (Strong Entity)

Three candidate keys: staff_id (PK), employee_code (UNIQUE), and email (UNIQUE). warehouse_id is a nullable FK — its nullability indicates central staff and does not create an intra-table transitive dependency.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| staff_id | → | {employee_code, full_name, role, email, phone, warehouse_id, is_active, last_login} | PK |
| employee_code | → | {staff_id} (+ all attributes via staff_id) | Alternate CK — UNIQUE constraint |
| email | → | {staff_id} (+ all attributes via staff_id) | Alternate CK — UNIQUE constraint |

Candidate Keys: {staff_id}, {employee_code}, {email}. Prime attributes: staff_id, employee_code, email.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values... | ✓**PASS** — 1NF holds. |
| 2NF | No partial dependency... | ✓**PASS** — 2NF holds. |
| 3NF | No transitive dependency... | ✓**PASS** — 3NF holds. |
| BCNF | LHS is a superkey. | ✓**PASS** — BCNF holds. |

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### REORDER_ALERT (System-Managed (Trigger-Populated))

Trigger-populated table. Single surrogate PK alert_id. reorder_point is a snapshot value captured at the time of alert generation.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| alert_id | → | {product_id, warehouse_id, preferred_supplier_id, current_stock, reorder_point, suggested_order_quantity, alert_status, triggered_at} | PK — only determinant |

### AUDIT_LOG (Append-Only Log (Trigger-Managed))

Immutable audit log. Single BIGSERIAL PK log_id. old_value and new_value are JSONB blobs — opaque atomic values, not repeating groups.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| log_id | → | {staff_id, action_type, table_name, record_id, old_value, new_value, performed_at, ip_address} | BIGSERIAL PK — only determinant |

### INVENTORY_MOVEMENT_LOG (Append-Only Log (Trigger-Managed))

Immutable stock movement ledger. Single BIGSERIAL PK movement_id. batch_id and zone_id are stored as historical snapshots for accuracy.

| Determinant (LHS) | → | Dependent (RHS) | Note / Justification |
|---|---|---|---|
| movement_id | → | {inventory_id, movement_type, quantity_change, quantity_after, batch_id, zone_id, reference_id, reference_table, performed_by, performed_at} | BIGSERIAL PK — only determinant |

**Note:** All three system-managed tables satisfy 1NF, 2NF, 3NF, and BCNF. Snapshot columns prevent live transitive dependencies.

### Proof for all three tables:

Candidate Keys: {alert_id}, {log_id}, {movement_id}, one each respectively for the tables given above. Prime attributes: alert_id, log_id, movement_id, respectively.

| Normal Form | Requirement | Verdict & Detailed Reasoning |
|---|---|---|
| 1NF | Atomic values... | ✓**PASS** — 1NF holds. |
| 2NF | No partial dependency... | ✓**PASS** — 2NF holds. |
| 3NF | No transitive dependency... | ✓**PASS** — 3NF holds. |
| BCNF | LHS is a superkey. | ✓**PASS** — BCNF holds. |

## NORMALIZATION SUMMARY

### Table-Wise Statistics

| Table | 1NF | 2NF | 3NF | BCNF | Notes |
|---|---|---|---|---|---|
| SUPPLIER | ✓ | ✓ | ×* | ×* | 3NF theoritically violated, practically achieved; BCNF violated by generated col. Retained for KPI performance. |
| PRODUCT_CATEGORY | ✓ | ✓ | ✓ | ✓ | BCNF achieved. Recursive self-ref FK is an attribute. |
| PRODUCT | ✓ | ✓ | ✓ | ✓ | BCNF achieved. |
| SUPPLIES | ✓ | ✓ | ✓ | ✓ | BCNF achieved. |
| WAREHOUSE | ✓ | ✓ | ✓ | ✓ | BCNF achieved. |
| WAREHOUSE_ZONE | ✓ | ✓ | ✓ | ✓ | BCNF achieved. |
| BATCH | ✓ | ✓ | ✓ | ✓ | BCNF achieved. |
| INVENTORY | ✓ | × | × | ×* | 1NF achieved. 2NF, 3NF, BCNF Violated – In a high-scale Supply Chain system, the cost of data redundancy is often lower than the CPU cost of performing multiple Joins and repeated arithmetic calculations on every read request.. |
| INVENTORY_ ALLOCATION | ✓ | ✓ | ✓ | ✓ | BCNF achieved. |
| PURCHASE_ORDER | ✓ | ✓ | ✓ | ✓ | BCNF achieved. GRN merge justified. |
| PO_ITEM | ✓ | ✓ | ×* | ×* | 3NF technically violated by generated line_total. BCNF violated. Retained as audit snapshot. |
| CUSTOMER | ✓ | ✓ | ✓ | ✓ | BCNF achieved. outstanding_balance NOT stored. |
| SALES_ORDER | ✓ | ✓ | ✓ | ✓ | BCNF achieved. |
| SO_ITEM | ✓ | ✓ | ×* | ×* | Same analysis as PO_ITEM. Retained as audit snapshot. |
| SHIPMENT | ✓ | ✓ | ✓ | ✓ | BCNF achieved. |
| DRIVER | ✓ | ✓ | ✓ | ✓ | BCNF achieved. |
| SHIPMENT_ITEM | ✓ | ✓ | ✓ | ✓ | BCNF achieved. |
| RETURN_REQUEST | ✓ | ✓ | × | × | 3NF violated: return_id → so_id → customer_id. Deliberate denormalization. |
| RETURN_ITEM | ✓ | ✓ | ✓ | ✓ | BCNF achieved. |
| STAFF | ✓ | ✓ | ✓ | ✓ | BCNF achieved. |
| REORDER_ALERT | ✓ | ✓ | ✓ | ✓ | BCNF achieved. |
| AUDIT_LOG | ✓ | ✓ | ✓ | ✓ | BCNF achieved. |
| INVENTORY_ MOVEMENT_LOG | ✓ | ✓ | ✓ | ✓ | BCNF achieved. |

* = violation documented and retained with explicit design trade-off justification.

### Aggregate Statistics

| Category | Count | Tables |
|---|---|---|
| BCNF achieved (fully normalized) | 18 of 23 (incl. 3 DDL Log Tables) | Product_Category, Product, Supplies, Warehouse, Warehouse_Zone, Batch, Inventory_Allocation, Purchase_Order, Customer, Sales_Order, Shipment_Item, Return_Item, Staff, Reorder_Alert, Audit_Log, Inventory_Movement_Log, Driver, Shipment |
| 3NF technically violated | 3 | PO_Item, SO_Item, Supplier |
| 3NF violated, deliberate cross-table denormalization | 1 | Return_Request |
| 2NF violated | 1 | Inventory |

All violations are explicitly documented, formally classified, and retained with justified design trade-offs.